Introduction to

Data Science & Machine Learning

From raw data to intelligent predictions

Session Agenda

01

What is Data Science?

Definitions, AI/ML/DS ecosystem & the DS process

02

Predictive Analytics

Types of analytics, PA workflow & applications

03

How Do We Make Predictions?

Benchmarks, rule-based models & ML comparison

04

Introduction to Machine Learning

ML types, algorithms, applications & workflow

05

Tools & Ecosystem

Python, libraries, platforms & resources

06

Roadmap & Q&A

Your learning path & open discussion

What is Data Science?

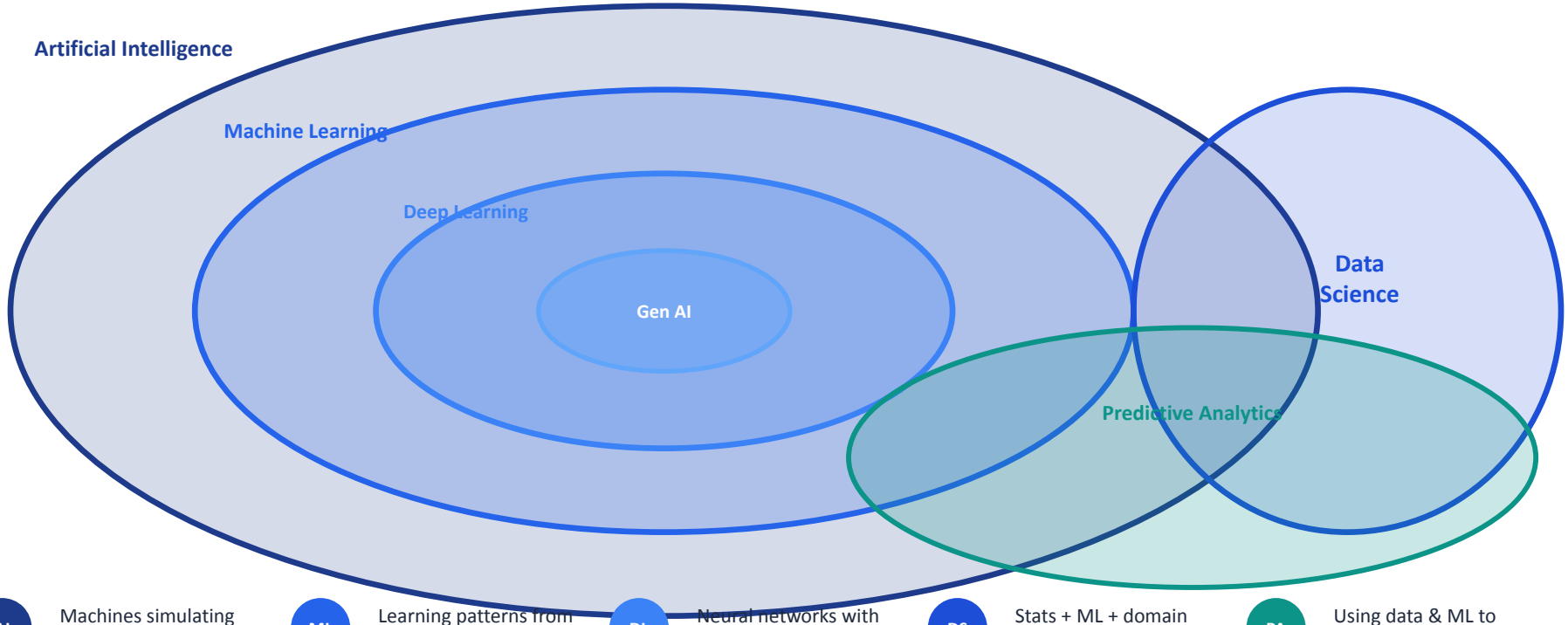
> *"Data Science is the art of extracting actionable insights from large, complex datasets using statistics, programming, and domain expertise."*

∑ Statistics & Math		</> Programming & CS		★ Domain Expertise
Probability, distributions, hypothesis testing	+	Python, SQL, algorithms, data structures	+	Business context, industry knowledge

= DATA SCIENCE

AI · ML · Deep Learning · Data Science · Predictive Analytics

These terms are related but distinct — here's how they nest and intersect:



AI Machines simulating human thinking

ML Learning patterns from data

DL Neural networks with many layers

DS Stats + ML + domain expertise

PA Using data & ML to predict outcomes

Predictive Analytics sits at the intersection of ML and Data Science — using algorithms to forecast future outcomes from historical data.

The Data Science Process

1	2	3	4	5	6	7
Problem Definition	**Data Collection**	**Data Cleaning**	**EDA**	**Modeling**	**Evaluation**	**Deployment**
Frame the business question	Gather raw data from sources	Handle missing/noisy data	Visualize & find patterns	Train ML algorithms	Measure model performance	Put model into production

This is an iterative process — you'll often cycle back to earlier steps as you learn more about the data

Types of Data

⊞ Structured Data

- Spreadsheets & CSV files
- SQL databases
- Financial transactions
- Sensor readings
- Survey responses

e.g. Age: 25 | Income: $60K | Churn: Yes

❖ Unstructured Data

- Text & social media posts
- Images & videos
- Audio recordings
- Email & documents
- Web pages (HTML)

e.g. "I love this product! 5 stars ★★★★★"

Semi-Structured: JSON · XML · HTML · Server Logs · CSV with headers · Email with metadata

Has some structure but doesn't fit neatly into rows & columns

What is Predictive Analytics?

Predictive analytics is the process of using data, statistical algorithms, and machine learning techniques to analyze current and historical data to make predictions about future events or behaviors.

Turns historical patterns into future insights

Types of Analytics

Descriptive Analytics

- Analyzes historical data to understand changes that have occurred
- Provides insight into past performance and reasons behind success or failure
- Uses Business Intelligence (BI), dashboards, and data visualization
- Example: Monthly sales reports showing historical sales trends

Predictive Analytics

- Uses statistical models and forecast techniques to predict outcomes
- Forecasts future probabilities and trends based on historical data
- Employs statistical algorithms and machine learning models
- Example: Predicting customer churn based on engagement metrics

Prescriptive Analytics

- Goes beyond predicting — suggests actions to benefit from predictions
- Advises on possible outcomes and recommends actions
- Uses simulation algorithms, optimization, and real-time decision-making
- Example: Logistics company optimizing routes to minimize costs

How Does Predictive Analytics Work?

1. **Data Collection**
   Gather relevant data from various sources

2. **Data Cleaning**
   Remove inconsistencies and prepare data

3. **Data Analysis**
   Explore and understand patterns in data

4. **Deployment**
   Deploy the trained model into production

5. **Model Monitoring**
   Continuously monitor and improve performance

Applications of Predictive Analytics

Business

Forecasting sales, optimizing marketing campaigns

Healthcare

Predicting disease outbreaks, personalizing treatment plans

Finance

Assessing credit risks, detecting fraudulent transactions

Sports

Enhancing team performance, predicting injury risks

Benefits of Predictive Analytics

- **Improved Decision Making** — More data-driven and informed decisions
- **Increased Efficiency** — Streamline processes and reduce waste
- **Enhanced Risk Management** — Identify and mitigate potential risks early
- **Competitive Advantage** — Stay ahead by predicting trends before they happen

How Do We Make Predictions?

Selecting Benchmark Models

- **Mean/Median Based Models** — Use central tendency for baseline predictions
- **Rule Based Models** — Create prediction rules from data analysis and domain knowledge
- **Machine Learning Models** — Algorithms that learn patterns from data automatically

Rule Based Models

Creating a set of rules to make predictions based on detailed data analysis and business understanding. Rules are manually crafted by domain experts.

Example: If Weather=Sunny & Humidity=Normal → Play=Yes

Rule Based System vs Machine Learning

Rule Based System

- Data ⟶ Human Analysis ⟶ Rules
- Humans craft prediction rules
- Works well for simple problems
- Limited by human understanding

Machine Learning

- Data ⟶ Algorithm ⟶ Rules
- Machine learns rules automatically
- Excels at complex problems
- Scales with more data

What is Machine Learning?

> ***"Instead of writing explicit rules, we give the machine DATA and let it LEARN the rules itself."***

Traditional Programming	VS	Machine Learning
Input Data		Input Data
Hardcoded Rules		Expected Output
→ Output		**→ Learns Rules!**
• *You manually write every rule*		• *Machine discovers patterns automatically*

Advantages of Machine Learning

- **Adaptability & Speed** — Quickly analyzes large amounts of data
- **Scalability** — Efficiently processes information as datasets grow
- **Consistency & Objectivity** — No fatigue or human biases in predictions

Applications of Machine Learning

Smart Navigation

Real-time traffic data analysis and route optimization

Finance

Fraud detection, risk management, and creditworthiness prediction

Marketing

Customer data analysis to improve marketing strategies

Education

Personalized learning by assessing student strengths and weaknesses

Agriculture

Optimize crop yields, predict crop health, recommend actions

E-commerce

Enhanced customer experiences with personalized suggestions

Real-World Applications

Healthcare

Cancer detection, drug discovery, patient readmission prediction

Finance

Fraud detection, credit scoring, algorithmic trading

E-Commerce

Product recommendations, dynamic pricing, demand forecasting

Transport

Self-driving cars, route optimization, traffic prediction

Social Media

Content moderation, feed ranking, ad targeting

Agriculture

Crop yield prediction, pest detection, precision farming

Types of Machine Learning

Three Main Types of Machine Learning

1. **Supervised Learning**
   Learning with labeled data — target variable is known

2. **Unsupervised Learning**
   Learning without labels — finding hidden patterns

3. **Reinforcement Learning**
   Learning through trial and error with feedback

Supervised Machine Learning

In supervised learning, there is a dependent or Target Variable, and the goal is to predict the target variable based on input or independent variables.

The algorithm learns from labeled training data

Supervised ML: Two Categories

Classification

- Target variable is categorical
- Predict spam email
- Predict movie genre
- Predict customer churn

Regression

- Target variable is continuous
- House price prediction
- Customer lifetime value
- Predict future temperature

Unsupervised Machine Learning

No Target Variable to predict. It focuses on finding hidden patterns and relationships within the data and grouping observations with similar patterns.

Discovers structure in data without predefined answers

Types of Unsupervised Learning

- **Clustering** — Organize similar objects into groups
- **Association Rules** — Find interesting relations among features
- **Dimensionality Reduction** — Reduce features while retaining important information

Reinforcement Machine Learning

Algorithms make decisions based on feedback from the environment. They follow a trial and error approach — receiving rewards for desirable actions and penalties for undesirable ones.

Feedback Loop: Action → Environment → Reward/Penalty → Learn

Spectrum of Machine Learning

From Structure to Freedom

- Supervised ML: Answers known in advance
- Unsupervised ML: Searching for hidden patterns
- Reinforcement ML: Continually adjusting based on feedback

Key Differences

- Supervised: Labeled data required
- Unsupervised: No labels needed
- Reinforcement: Environment interaction

Key ML Algorithms — Big Picture

(We'll deep-dive each in future sessions)

Linear Regression
Supervised

Predict continuous values (price, score)

Logistic Regression
Supervised

Binary classification (yes/no)

Decision Trees
Supervised

Rules-based classification & regression

Random Forest
Supervised

Ensemble of trees — high accuracy

K-Means Clustering
Unsupervised

Group similar data points together

SVM
Supervised

Find optimal decision boundary

Neural Networks
Deep Learning

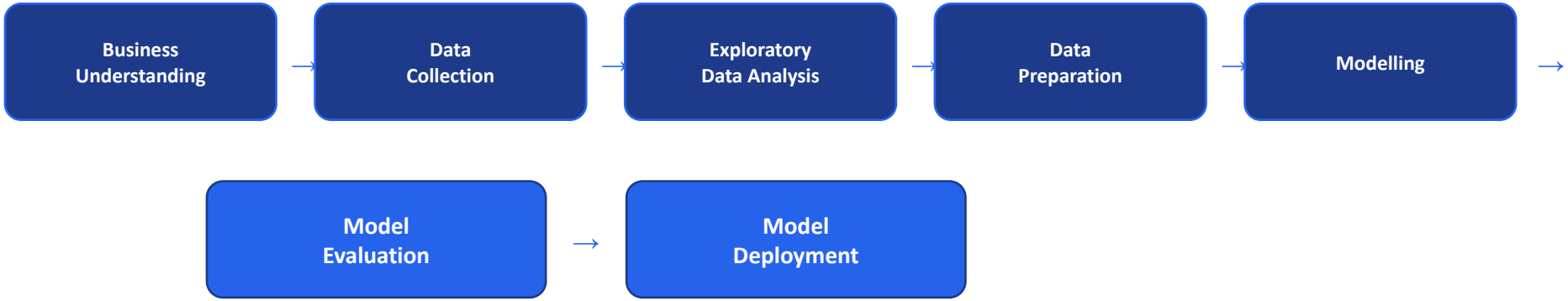
Complex pattern recognition (images, NLP)

K-Nearest Neighbors
Supervised

Classify by similarity/proximity

Machine Learning Workflow

Machine Learning Workflow



Each step feeds into the next — forming an iterative, data-driven pipeline.

Data Preparation

1. **Data Division**
   Split into train and test data for reliable evaluation

2. **Data Cleaning**
   Remove invalid values, impute missing values, treat outliers

3. **Data Transformation**
   Convert data from one format to another

4. **Data Restructuring**
   Combine data from multiple sources

5. **Feature Engineering**
   Create new features by combining existing ones

Modelling & Evaluation

- Train different ML algorithms using the prepared dataset
- **Regression Models** — Predict continuous values
- **Classification Models** — Predict categorical outcomes
- **Unsupervised ML Models** — Find hidden patterns
- Evaluate with metrics: MAE, RMSE, Accuracy, Precision, Recall

Model Deployment

The trained and evaluated model is deployed or integrated into the ecosystem. This is the final step where the model starts making real predictions.

From data to decisions — the ML pipeline comes full circle

Tools & Ecosystem

The Python data science stack — everything you need to get started

Language	Data Manipulation	Visualization
Python Primary DS/ML language	**Pandas** DataFrames & wrangling	**Matplotlib** Static charts & plots
R Statistical computing	**NumPy** Numerical arrays	**Seaborn** Statistical viz

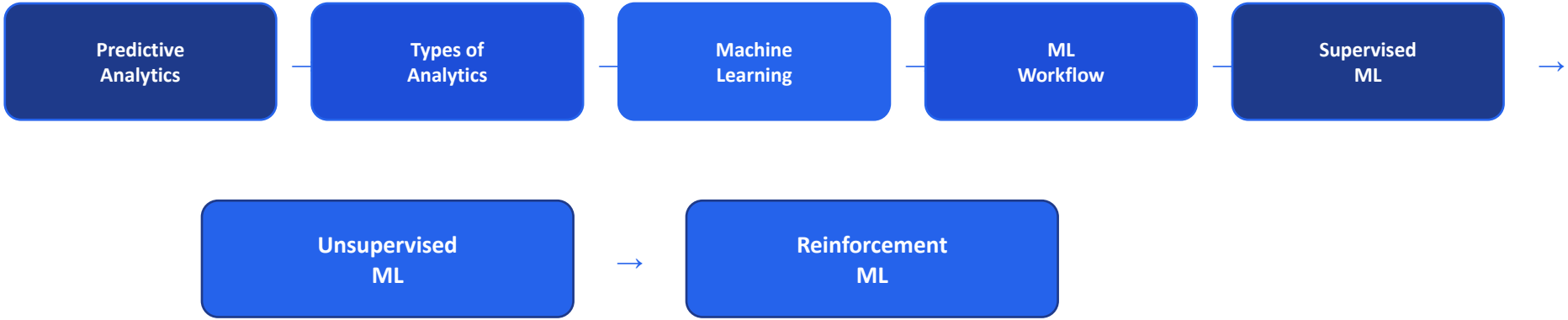
ML / Deep Learning Libraries

Scikit-learn	TensorFlow	PyTorch	Jupyter
Classical ML	Deep learning	Research DL	Interactive notebooks

Platforms:

Google Colab · Kaggle · Databricks · AWS SageMaker · Azure ML · Hugging Face

Key Takeaways



Each concept builds on the last — together they form the complete data science toolkit.

What's Coming Next

Session-by-session breakdown of this course

Linear & Logistic Regression — Supervised

Decision Trees & Random Forests — Supervised

K-Means & Clustering — Unsupervised

Neural Networks & Deep Learning — Deep Learning

NLP & Text Classification — NLP

Model Evaluation & Tuning — MLOps

End-to-End Capstone Project — Project

Today we planted the seed. Each session, you'll go deeper. Let's build something amazing together! 🚀

Continue Learning

This was an introduction to Data Science, Predictive Modeling, and Machine Learning. In the next sections, we'll dive deeper into each algorithm type.

Stay curious. Keep learning. Build the future. 🚀

Questions & Discussion

Open floor — ask anything!

Next: Linear & Logistic Regression →

What we covered today:

- ✓ What Data Science is and why it matters
- ✓ The end-to-end data science process
- ✓ Machine Learning fundamentals
- ✓ Supervised · Unsupervised · Reinforcement
- ✓ Key algorithms overview
- ✓ The Python tools ecosystem